FOR THE WHOLE TEAM

Kanda Rakshak in simple words

How it works, from the sensor in the onion pile to the fan to the message on the farmer's phone. Read this once and you can explain it to anyone.

Kanda Rakshak is a kit that goes into the onion shed a farmer already has. It runs the fans at the right time to keep the onions dry, warns early when the pile is getting into trouble, and tells the farmer in Marathi whether to hold, sort or sell.

The problem, in 4 lines

- After harvest, farmers store onions for 3–5 months in a shed called a **kanda chawl**, hoping prices go up.
- Onions are alive. They breathe and give off heat and water. If the pile gets too wet or warm inside, onions at the **bottom start rotting**.
- Nobody can see it. By the time the farmer smells it, a big part is gone. Up to **40–60%** can be lost.
- He also doesn't know **when to sell**. Too early, low price. Too late, rotten onions.

Think of the kit like a body

THE SKIN

Temperature + humidity sensors

Feel **heat and wetness**. Three sit inside the pile on a stick (top, middle, bottom), one outside in the shade. PoC: DHT22. Field kit: SHT40 (more stable when very humid) plus a **temperature string** through the heap to catch hot spots.

THE NOSE

Gas sensors

Smell trouble early. PoC: MQ-135/136 for sulphur and ammonia. Field kit: a **CO_2 sensor combined with a sulphur sensor** (BME688, or MQ-136 where there is grid power). CO_2 rises when onions breathe harder; sulphur says whether it is rot. Gas only confirms; it never decides alone.

THE BRAIN

ESP32

A ₹450 mini-computer. It **reads all the sensors** every few minutes, **decides the fan** by itself, and **sends the readings** to our server.

THE LUNGS

Fans + switch

Fans **push air into the channel under the floor** so it rises through the pile. A 5 t shed needs one fan; a 25 t chawl needs 2–3 bigger (300 mm) fans. A relay or solid-state switch lets the ESP32 turn them on.

THE HEART

Power

A small **solar panel + battery** runs the electronics day and night. Big fans run on **farm power**; fully off-grid fans need a much bigger panel and battery. The buck converter turns 12 V into 5 V for the electronics.

THE MOUTH

Our server + phone message

Our server works out the risk, checks the onion price, and **tells the farmer one instruction in Marathi**.

Safety part you'll hear about: a **voltage divider** (two resistors) makes sure the gas sensors' 5 V signal is reduced to 3.3 V, because more than 3.3 V would burn the ESP32.

The journey of one reading, step by step

1. **Feel and smell.** Every few minutes, the sensors measure heat and humidity at the top, middle and bottom of the pile and outside, and the gas sensors check for rot smell.
2. **Decide the fan, right there in the shed.** The ESP32 asks: “Is the pile too wet or warm, and will the outside air actually dry it without forming dew on the onions?” If yes, the fans turn ON. On a foggy or monsoon night they stay OFF. **This needs no internet.** (Field kit uses the dew-point version of this rule.)
3. **Send the reading.** The ESP32 sends the numbers to our server over the internet (Wi-Fi in the demo, a 4G SIM in the field). If there's no network, it **keeps the readings in memory** and sends them later.
4. **Give it a score.** The server turns the readings into one **rot-risk score from 0 to 100**: below 35 is SAFE, 35–65 is WATCH, 65 and above is ROT.
5. **Check the market.** The server checks today's onion price at **Lasalgaon mandi** from the Government's own website (data.gov.in). It shows the date and source every time.
6. **Pick one action.** Risk + price → one instruction: **Hold**, **Sort** or **Sell**.
7. **Tell the farmer.** He gets one short message in Marathi, as text and voice. In our demo it comes on **Telegram**. In the real product it will be an **SMS and a phone call**, which work on any basic phone.

How the risk score works (with a real example)

Three things are checked and added up, each out of 100:

What we check	Why it matters	Weight	Example: rotting shed
Rot gas	Earliest direct sign of rot	50%	Gas is 80% above the shed's fresh level → 100 points → 50
Humidity	Onion stores best at 65–70%. Above 80–85%, rot spreads fast	30%	Bottom layer 91% → 100 points → 30
Heat	A hot spot inside the pile means rot or heavy breathing	20%	Bottom is 3.7°C warmer than outside → 92.5 points → 18.5

Total: 50 + 30 + 18.5 = 98.5 → ROT. The farmer is told to sort today.

This is version 1 of the score. Version 2 adds heap hot spots (temperature string) and how fast CO_2 rises, and the pilot will tune the weights.

Why the gas sensor compares to the shed's own “fresh” level: cheap gas sensors aren't exact, but they're very good at noticing a **change**. So when fresh onions are loaded, we remember that reading and watch for it to rise.

The four messages the farmer can get

Situation	Message (Marathi)	Meaning
Onions are safe, price normal	सुरक्षित · थांबा	Hold. Don't sell yet.
Safe, but price is 10%+ higher than last week	सुरक्षित · विकण्याचा विचार करा	Consider selling some.
Getting too wet or warm	लक्ष ठेवा · तपासा	Check the pile; the fan is handling it. (If price is rising: sell this week.)
Rot has started	धोका · निवडणी करा	Danger: sort today and remove the bad onions.

Messages are fixed sentences written by us in Marathi and checked by native speakers. No AI translation, so the advice is always exact. A new message is only sent when the advice changes and stays the same for 3 readings, so the farmer is never spammed.

Which gas comes first when onion rots?

Order	What happens	Gas
1st	Fungus or bacteria start attacking	Sulphur compounds (the rotten-onion smell), a little H_2S
2nd	Onions breathe harder and warm up	CO_2 rises, plus heat
3rd	Tissue starts fermenting	Alcohols and other VOCs
4th (late)	Onion turns mushy	Ammonia

CO_2 + sulphur together tell the cause: CO_2 and heat up but no sulphur = too warm or sprouting (run the fan). CO_2, heat *and* sulphur up = rot (sort today). Other crops work the same way: they all breathe harder when going bad, plus their own smell (grain: musty mould; fruit: alcohol).

Who pays and how we earn

- **Pay per tonne, not per kit:** FPOs pay a small fee per tonne stored per season; we own and maintain the kit.
- **Same kit, many crops:** onion, then potato, garlic or grain storage.
- **Subsidy:** a 25 t chawl gets up to ₹1.25 lakh under MIDH; our kit is about 6–8% of a chawl's cost.
- **Hold with credit:** verified stock can back a warehouse-receipt loan, so the farmer doesn't have to panic-sell.
- **From harvest day:** curing reminders and a loading checklist cut loss before the onion even enters the shed.

What happens with no internet?

- **The fan keeps working.** The decision is made inside the shed by the ESP32.
- **Nothing is lost.** Readings are saved and uploaded when the network comes back.
- **No wrong alerts.** Old readings are marked as old, so they don't trigger a fresh alert.
- **The farmer needs no internet.** SMS and phone calls work on a basic phone without mobile data.
- One-line answer: **“Protecting the onions never needs the internet. Only sending advice does.”**

What is real today, and what is next

Part	Status
Server, risk score, advice, Marathi messages	**Built and tested** (79 automated tests)
Live onion price from the Government (data.gov.in)	**Working**
Message on a phone	**Working** on Telegram (demo). SMS + voice call: **planned**
Farmer view, shed view, department view	**Built**
Digital twin (8 pretend sheds around Nashik)	**Running**. Always labelled **SIMULATED**
Circuit diagram	**Finalised**
ESP32 program (firmware)	**Written**, not yet tested on the board
Bench proof of concept	**Next**: ₹11,500, about 4 weeks
Field kit v2 (SHT40, CO_2, temperature string, bigger fans, dew-point rule)	**Designed and costed**, built after the PoC
Pilot	**Planned**: 40 kit sheds in Niphad–Lasalgaon, compared with 20 normal sheds

Key numbers

What	Number
Bench proof of concept	₹11,500
Field kit, 25 t chawl (fans included)	≈ ₹17,000 (≈ ₹700 per tonne)
Small 5 t shed	≈ ₹9,000–10,000
Proof that ventilation works	NHRDF Sinnar: 14.5% loss with fans vs 33.4% without (6 months)
Subsidy for a 25 t chawl (MIDH 2025-26)	Up to ₹1.25 lakh (50%)
Running cost per farmer	Messages under ₹60 per season; data SIM ₹600–1,200
Maharashtra's share of India's onion	40–45%
Stored every year	40–50 lakh tonnes
Lost in normal sheds	40–60% within about 4 months
Today's Lasalgaon price	₹_____ per quintal (check before the call)
Award	1st Prize, InnovateX 2.0, IEEE EMBS Pune × MIT-ADT (30 Oct 2026)

Words you'll hear

Word	Simple meaning
Kanda chawl	The farmer's onion storage shed, with slatted walls for air.
Quintal	100 kg. Mandi prices are per quintal.
Modal price	The most common price that day at the mandi.
Agmarknet / data.gov.in	Government websites where every APMC mandi publishes daily prices.
Baseline	The gas reading when onions are fresh. We watch for it to rise.
Digital twin	A computer copy of a shed that behaves like a real one, used for testing.
Relay	An electric switch. The ESP32 flips it to turn the fan on or off.
Buck converter	Turns 12 V from the battery into 5 V for the electronics.
FPO	Farmer Producer Organisation: a group of farmers who buy and sell together.
DLT	TRAI's registration system for business SMS in India.